Financial Data Analysis in Python

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***Abstract*— This report is about a mini project focused on exploring, analyzing, and modeling financial data using Python. The project involved univariate and bivariate exploratory data analysis (EDA), statistical analysis, and time series modeling on a financial dataset. The chosen dataset consisted of price, volume and market cap of a famous memecoin cryptocurrency Dogecoin.**

Keywords—python, dogecoin, cryptocurrency, financial, time series, EDA

I. Introduction

Financial Data Analysis using python project is all about finding insights and useful information about the financial data using several analysis technique such as EDA, Time Series Analysis, Statistical Tests etc. The objective of this mini project is to find the answers to below research questions and to get the idea of how python can be used to analyze, exploration and time series modeling on a financial data.

1. How did the historical price and market capitalization of the cryptocurrency changed over time?
2. Predicting the future prices of cryptocurrency?
3. Checking the volatility of Dogecoin over time?
4. What factor/season changes the price of the cryptocurrency?
5. Relationship between Market Capitalization and Price.
6. Relationship between Volume and Prices.

II. Methodology

The dataset used in this analysis was obtained from kaggle.com (an online dataset repository), containing information of famous meme coin cryptocurrency named "Dogecoin". The dataset consists of Price(High, Low, Open, Close), Volume and Market Capitalization of the cryptocurrency. All analyses and visualizations were created using python in google colab using modules such as seaborn, matplotlib, numpy, pandas, sklearn, scipy.stats etc.

This methodology facilitated a comprehensive exploration of the financial data mentioned above utilizing a variety of graphical techniques to uncover insights and patterns. Visualizations are generated using matplotlib and seaborn to effectively communicate the findings of the analysis.

The dataset consists of 4 features related to the price of the cryptocurrency with High, Low, Open and Close price of it for a particular date. The dataset also includes the volume of the cryptocurrency that has been traded everyday and the market capitalization on that day. The dataset has 2760 rows out of which 11 rows of Volume feature has null or zero values. The date feature is in MMDDYYYY – HHMM format, however the time in rows has an entry of 23.59 which is similar to every row which suggest that the data (price of the dogecoin) is collected at the very end of the day. Everyday price is a numerical value with floating point numbers, also the Volume and Marketcap are in floating point numbers. The other three columns are SNo which is a counter of the day, the column Name and Symbol are similar for every row and is not going to play any role in the analysis of the price. The data is continuous from 16/12/2013 to 7/6/2021 which will be used for time series analyses of the price of the dogecoin.

III. Analysis

The dataset is able to justify the needs and useful in finding the answers of research questions mentioned. Several operations and statistical technique has been applied on the dataset in order to find relationship between features. Univariate and Bivariate analysis, EDA, Statistical Analysis and Time Series analysis. The brief of these analysis are as follows:

1. **Univariate Analysis:** The Dataset contains several numeric features for the price and volume of the cryptocurrency, below is an snippet of a summary of the key descriptive statistics of the numeric variables:

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Basic Column wise Summary of the Database such as Mean, Median, SD etc is as Follows:

                SNo         High          Low         Open        Close  \
count  2760.000000  2760.000000  2760.000000  2760.000000  2760.000000
mean   1380.500000     0.014964     0.012402     0.013669     0.013763
std     796.887696     0.069103     0.055536     0.062347     0.062559
min       1.000000     0.000089     0.000085     0.000087     0.000087
25%     690.750000     0.000231     0.000222     0.000227     0.000226
50%    1380.500000     0.001914     0.001772     0.001849     0.001844
75%    2070.250000     0.002986     0.002796     0.002900     0.002899
max    2760.000000     0.737567     0.608168     0.687801     0.684777

             Volume     Marketcap
count  2.760000e+03  2.760000e+03
mean   4.326321e+08  1.760533e+09
std    2.807320e+09  8.115186e+09
min    0.000000e+00  1.509085e+06
25%    2.536138e+05  2.337892e+07
50%    5.582480e+06  2.135789e+08
75%    5.098244e+07  3.488812e+08
max    6.941068e+10  8.868082e+10
```

Fig 1.1

The Univariate analysis provides that most of the variables are positively skewed with values grouping near low ranges and occasionally spikes and peaks. The price of the cryptocurrency, volume and market cap shows high variability in the later period of time.

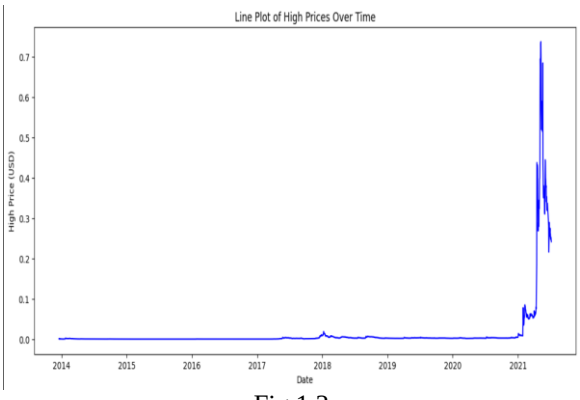

Fig 1.2

2. **Bivariate Analysis:** Data is visualized in several types of graphs and plots such as Scatter plot, Correlations Matrics, Heatmaps to explore bivariate relationship between numeric variables.

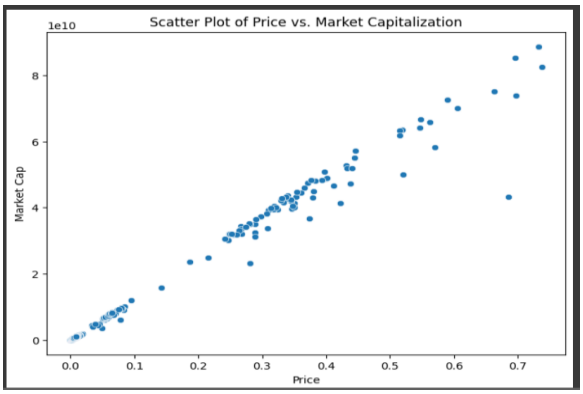

Fig 2.1

A Scatter plot has been generated to analyze relationships between the Price and Market Cap of the cryptocurrency. The figures shows that there is a positive correlation between the price and the Market cap. As the price of cryptocurrency increases, its market capitalization generally also increases. The datapoints are not uniformly distributed, there is a higher concentration of points towards lower end of the price range.

3. **Correlation Matrics:** A correlation matrix has been plotted between the variables High, Low, Market Cap and Volume. Value ranges from 0 to 1 where 0 represent no correlation and 1 represent perfect positive correlation.

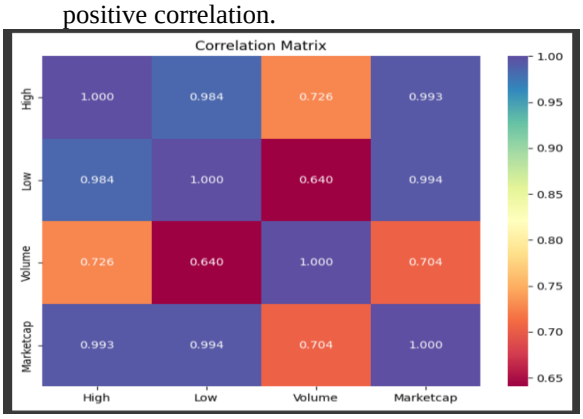

Fig 3.1

4. **Statistical Analysis:** The Statistical analysis in the form of several testes has been done on different features of the dataset including T-Test, ANOVA and Linear Regression to compare the mean of features with other aspects of the dataset. The results of the mentioned tests are briefly explained.

A. T-Tests: T Test is used to compare the means of two features and determine if there is a difference between those two features. T-tests has been performed in three aspects:

- One Sample Test: The One Sample test statistic measures the difference between sample mean and the hypothesis mean. The value of 0.733 means that the sample mean in 0.733 standard errors away from the hypothesis mean. The p-value is 0.464 which represents the probability is much higher than the typical value i.e 0.05, this suggests that there is not much of difference between the population mean and the sample mean.
- Two Sample Test: The results of t-statistic is: 1.51 with p-value being 0.129. The T test statistic suggest that two groups(High and Low) are not significantly different . The p-value 0.129 suggests that there is not much of difference between the High and Low values.
- Paired T test: Paired T Test between Open and Close has been done. T-statistic is –0.534 and the p-value is 0.593. The T-statistic value suggests that the difference between mean of Open and Close is very small. However, the negative sign shows that the value of Close is higher than the value of Open. The p-value being 0.534 suggest that the two columns are highly related to each others.

B. Analysis of Variance(ANOVA): The analysis has been done on High, Low and Open price of the crypto. The F-Statistic value is 0.77 and the P-Value is 0.509. This low F Statistic value suggest that the variance between the groups are not large enough, means there is not a significant difference between these three features. A P value of 0.77 suggest that there are not enough evidence to say that there are differences between means of High, Low and Open prices of the cryptocurrency.

5. **Time Series Decomposition:** Time-series decomposition technique has been used to break down a time series into its constituent components such as trends, seasonality, and residuals. Additive decomposition was used, assuming that the components add up to the observed data. A period of 30 days was chosen to capture monthly seasonality.

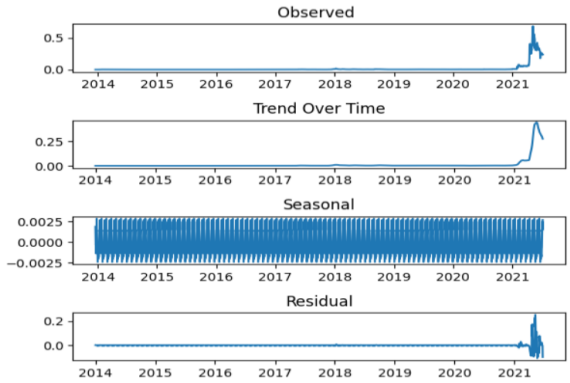

Fig 5.1

The "Trend Over Time" component shows a general up and down movement in Dogecoin prices over time. The seasonal component reveals recurring patterns, such as periodic price fluctuations. The residuals represent noise or irregular fluctuations that cannot be explained by trend or seasonality.

6. **Forecasting Model (ARIMA)**: ARIMA (Autoregressive Integrated Moving Average) is a widely used statistical model for time-series forecasting. The data was split into training (80%) and testing (20%) sets. An ARIMA model with order (5, 1, 0) was fitted to the training data. Forecasts were generated for the test set. The model's performance was evaluated using MSE, measuring the average squared difference between actual and predicted values.

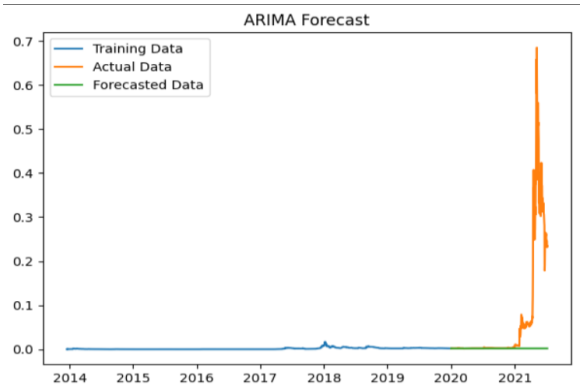

Fig 6.1

IV. Conclusion

The EDA reveals a strong positive relationship between price and market capitalization. The correlation matrix quantifies these relationships, showing strong correlations between price and market cap and moderate correlations between price, volume, and market capitalization. While analyzing the Univariate most of the variables are positively skewed with values grouping near low ranges and occasionally spikes and peaks. the price of the cryptocurrency, volume and market cap shows high variability in the later period of time.

Time series analysis shows a comprehensive workflow for analyzing and forecasting Dogecoin prices using time series techniques. Major steps including preprocessing the data, decomposing it into trend, seasonality, and residuals, and evaluating the performance of ARIMA and moving average models. While ARIMA provided better accuracy, the moving average model offered simplicity and ease of interpretation. The ANOVA test indicates no significant differences between the High, Low, and Open prices, while the t-test helped assessing differences between two specific groups. Combining these tests provide valuable insights into the relationships and significance of variables in the dataset.

Acknowledgments

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